A RECONSTRUCTION PROGRAMME IN STRUCTURAL MATHEMATICS

# UNIFIED MECHANICS

## COMPACT MONOGRAPH

FINAL EDITION · 2026

*From local records to collective fields,*
*without replacing established physics.*

$\varphi$

$$(R_{n+1}, C_{n+1}) = (R_n + C_n,\ R_n) \implies \varphi^2 = \varphi + 1, \qquad r = \frac{1}{2\varphi} = \frac{\sqrt{5}-1}{4}$$

JOSEPH SHIELDS

14 July 2026

ONE INVARIANT · MANY SECTORS · ONE CONSISTENCY TEST

**Status.** Internally complete reconstruction framework; four explicit physical anchors and one dimensional calibration; ensemble interpreter implemented and executed on an accepted cosmological history; TNG literature phase complete (language survives; superiority untested); TNG catalogue test execution-ready but deferred; SPARC pilot complete with an honest baseline-equivalent null result; optional physical hypothesis frozen; independent empirical verdict open.

# Contents



*The mathematics can be the exact language of what physical records collectively describe, without becoming a new substance.*

# EXECUTIVE SUMMARY

Unified Mechanics (UM) is best understood as a disciplined language for how physical records become useful descriptions, together with a finite, explicit proposal for the mechanism those descriptions share. One atom can produce an atomic record. A sea of atoms can support density, temperature, pressure, composition, velocity, and radiation fields, because many local records form stable ensemble statistics. The field is not a substance added to the atoms; it is what the mathematics describes at a declared scale.

UM therefore has four layers, and the final edition never blends them:

1. a **structural grammar** of update, relation, retention, normalization, quotienting, and internal age — genuinely derived mathematics;
2. an **observational language** of scale-dependent Boundaries, records, Transmission, Familiarity, and four clocks;
3. an **ensemble compiler** that reinterprets the outputs of accepted physics without changing the physical solver;
4. an **optional density-gated scalar hypothesis**, tested separately and allowed to fail.

Between layers 1 and 3 sits the exact physical boundary of the programme: four representation anchors (A1–A4) and one dimensional calibration, $S_* r^3 = \hbar$. Everything physical in UM is conditional on that finite premise set, and nothing else.

THE NON-NEGOTIABLE RULE

**No mechanism → no derived observable → no physical prediction.** A measurement exists only after the framework specifies what is measured and how it becomes a record:

physical world → interaction → channel → observable → record → calibration → inference

The universe does not need a global variable stating that a person is talking to an AI. The conversation exists through a local causal chain and persists in the records available to compatible receivers. A cosmological description can omit that fine detail while the event remains locally real.

The central empirical question is:

*Does a fixed UM compiler transfer between systems, epochs, and resolutions, and predict withheld records better than generic coarse-graining?*

The 2026 empirical phase answered part of that question honestly: the UM language survived a nine-paper IllustrisTNG literature comparison without contradiction but without established superiority; the frozen TNG catalogue test is execution-ready and deferred on access and the exact adapter; and a real-data SPARC pilot executed the quadratic compiler on twelve galaxies and returned a baseline-equivalent null result. Until a distinctive test is passed on independent data, UM is a coherent and executable reconstruction framework, not a validated replacement for established physics.

# 1 WHAT KIND OF CLAIM IS BEING MADE?

Every claim in the programme carries exactly one grade. Elegance and numerical similarity do not change a claim's grade; relabelling a known result is interpretation, not physical confirmation.

| GRADE | MEANING | TYPICAL EXAMPLE |
|---|---|---|
| **D** | definition or declared notation | a Boundary or kernel definition |
| **T** | theorem under stated assumptions | a normalization or factorization result |
| **E** | established external science | recombination physics, stellar nucleosynthesis |
| **R** | calibrated record from observation or simulation | a measured spectrum, a saved simulation field |
| **U** | UM interpretation or model postulate | assigning Light/Matter coordinates to records |
| **H** | optional new-dynamics hypothesis | the density-gated scalar response ratio |
| **Q** | quarantined legacy claim | an unbridged numerical coincidence |

A theorem inherits every condition in its dependency chain. A physical statement whose path contains an open realization map remains open. The exact verbs matter: UM *derives*, *realizes*, *reconstructs*, *interprets*, *proposes*, *preregisters*, *predicts*, *tests*, *fails to distinguish*, or *remains open* — and never writes "UM predicts $X$" for what is only a retrospective mapping.

## 2 MINIMUM STRUCTURAL CORE

The internally closed whole-shift relation (Axiom A) is

$$(R_{n+1},\, C_{n+1}) = (R_n + C_n,\; R_n),$$

whose encoder $F = \left(\begin{smallmatrix}1 & 1\\ 1 & 0\end{smallmatrix}\right)$ has the unique positive fixed ray $\varphi^2 = \varphi + 1$. PROVED Within that selected recursion and an explicit Normalization $N$, UM uses the contraction coordinate

$$r = \frac{1}{2\varphi} = \frac{\sqrt{5}-1}{4} \approx 0.3090169944, \qquad u = 1 - r \approx 0.6909830056, \qquad 4r^2 + 2r - 1 = 0.$$

These are exact structural coordinates under Axiom A and $N$. Generic self-description does not uniquely force the recursion NO-GO, and the coordinates are not automatically measured densities, couplings, or cosmological parameters.

For normalized local amplitudes $a_L + a_M = 1$, the quadratic channel grammar is

$$W_L = a_L^2, \qquad W_B = 2a_L a_M, \qquad W_M = a_M^2, \qquad W_L + W_B + W_M = 1.$$

The exact identity separates update-like, mixed-Boundary, and retention-like terms. PROVED Physical meaning begins only after the amplitudes are constructed from declared records; the identity does not prove that nature uniquely prefers these coordinates, and three channel types do not by themselves imply three spatial dimensions — that requires the conditional survival theorem of the main monograph.

The observation chain is typed:

$$\text{source state} \xrightarrow{M_S} \text{physical record} \xrightarrow{C_S} \text{familiar class} \xrightarrow{O_S} \text{reconstructed invariant}$$

A quotient can remove redundant descriptions; it cannot recreate a distinction that never reached the record. Internal age is an additive cocycle on a declared retained history PROVED; one scalar age does not uniquely determine geometry, spacetime dimension, or a physical field NO-GO.

## 3 The Physical Boundary: Four Anchors, One Calibration

The exact boundary between the formal reconstruction and any claim about Nature is a finite, visible premise set. Nothing physical is claimed outside it, and no sector may quietly retune it.

| Anchor | Physical meaning | Why it matters |
|---|---|---|
| **A1** | Nature realizes the whole-shift referent/code update physically, not only formally. | Connects the golden fixed ray to a physical update process. |
| **A2** | Formal independence and local execution are represented by a conservative local monoidal (continuum) system. | Lets the abstract composition rule become local physical dynamics. |
| **A3** | One relational registration machine converts execution throughput into records and stress-energy. | Connects channel routing to cosmological and vacuum observables. |
| **A4** | The complete three-coordinate outcome algebra is carried by a closed spin compact internal structure. | Connects the channel algebra to a physical internal carrier. |
| **Cal.** | $S_* r^3 = \hbar$ | Supplies the dimensional quantum anchor that a dimensionless recursion cannot supply by itself. |

**The Right Physical Claim**

UM does not claim that its axiom directly emits every observation. It claims that the formal structure constrains a small family of possible physical mechanisms, and that concrete predictions arise only after a physical realization, coupling, readout, and calibration are specified.

## 4 Effective Objects and the Actual Boundary

An **effective object** is a metastable identity class at a declared scale. It can persist while matter, energy, and information cross its Boundary. The **actual Boundary** is the causal cut between retained identity and exchanged information for a selected scale, channel, and question.

One target can therefore have several nested Boundaries:

- a planet's visible surface, atmosphere, interior, magnetic region, and orbit;
- a star's photosphere, nuclear interior, wind, and gravitational system;
- a galaxy's disk, gas, halo, satellites, and environment;
- a black hole's emitting plasma, compact exterior, and relativistic horizon.

Unresolved variables can continue to influence retained variables, so a coarse description may contain memory even when a finer theory is local.

## 5 Transmission and Familiarity

For measurement map $M_S$ and validated decoder $C_S$ at scale and channel $S$, the **Transmission Kernel** is

$$K_S^T = \{\Delta X : \ M_S(X + \Delta X) = M_S(X)\}.$$

It contains world distinctions absent from the record. No decoder can recover them from an unchanged record.

The **Familiarity Kernel** is

$$K_S^F = \{d : \ C_S(d) = \text{unfamiliar}\}.$$

It contains structure that was recorded but does not yet belong to a validated language class. It may shrink when a new class is learned and independently checked.

Missing information and unfamiliar information are therefore different. For independent linear channels, $K_{\text{joint}}^T = \bigcap_i K_i^T$: adding a genuinely informative physical channel can shrink the joint Transmission Kernel. Adding a synonym cannot.

## 6 Four Ages, Gradients, and Complexity

Every age claim should name

$$\tau = (\tau_{\text{signal}}, \ \tau_{\text{formation}}, \ \tau_{\text{material}}, \ \tau_{\text{processing}}).$$

Signal age is travel time. Formation age dates assembly. Material age dates the constituents. Processing age dates the latest important reworking. A Jupiter image is minutes old as a signal, Jupiter is billions of years old as an object, many of its nuclei are older, and its atmosphere is changing now.

Distance creates propagation delay and dilution, not a universal temperature or complexity law. Temperature, pressure, chemical-potential, radiation, charge, and gravitational-potential gradients drive flows. Long-lived organization needs a source of free energy, a selective Boundary, retention, and recurring familiar records.

Complexity is a profile, not one ranking number for the whole universe:

$$\kappa = (\text{Diversity}, \ \text{Nesting}, \ \text{Retention}, \ \text{Channels}, \ \text{ModelDepth}, \ \text{Uncertainty}).$$

## 7 From One Atom to a Field

For records $z_i$ at positions $x_i$, a coarse variable at scale $\ell$ is

$$F_\ell(x) = \frac{\sum_i W_\ell(x - x_i)\, f(z_i)}{\sum_i W_\ell(x - x_i)}, \qquad N_{\text{eff}} = \frac{\left[\sum_i w_i\right]^2}{\sum_i w_i^2}.$$

At $N_{\text{eff}}$ near one, the value is essentially local. At large $N_{\text{eff}}$, with controlled uncertainty and stability under reasonable resolution changes, field language becomes operationally justified.

Means alone are insufficient. Fluctuations and correlations carry clustering, waves, line broadening, turbulence, and anisotropy. Different microscopic histories can produce the same retained collective field; a stable macroscopic description need not know every constituent's history.

A useful Boundary may occur where a population changes, a field becomes steep, a constitutive rule changes, two components become comparable, or an interaction rate crosses a dynamical rate. For non-negative records $a$ and $b$,

$$B_{\text{eq}}(a, b) = \frac{4ab}{(a + b)^2}$$

is one bounded equality marker. It labels a crossover already present in the records. It is not a new energy component.

## 8 EARLY UNIVERSE AS A RETENTION HIERARCHY

UM reinterprets established early-universe science without replacing it. INTERPRETATION U

**HOT EQUILIBRIUM**
Fast interactions repeatedly rewrite microscopic states. Conserved quantities, collective variables, distributions, and fluctuations are the durable records.

**FREEZE-OUT**
When an interaction rate falls below the expansion rate, a distribution stops tracking equilibrium. The relic abundance becomes a retained record of that crossing.

**MASS THRESHOLDS AND CONFINEMENT**
Species becoming nonrelativistic change their stress-energy trace. QCD confinement changes the useful language from quarks and gluons to hadrons; most ordinary nucleon mass belongs to the strongly bound state.

**NUCLEOSYNTHESIS**
Nuclear identities and light-element abundance records form through established reaction networks. A structural coordinate cannot replace those rates and initial conditions.

**RECOMBINATION**
Neutral atoms change photon transport and release the long-range cosmic microwave background record. This is a major transmission Boundary, not the first moment matter became real.

**STRUCTURE AND STARS**
Gravity produces density and potential gradients. Cooling and dissipation allow nested structures. Stars extend the nuclear language and distribute processed material into later gas, planets, chemistry, and life.

## 9 The Periodic Table as a Physical Language

For isotope and environment $(Z, A, e)$, use

$$\Gamma_{(Z,A,e)} = (E_{\text{levels}},\ \sigma_{\text{abs}},\ \sigma_{\text{emit}},\ \tau_{\text{levels}},\ B_{\text{nuc}},\ \alpha_{\text{pol}},\ \text{phase}).$$

Laboratories establish recurring transition identities; remote spectra reuse them to infer composition. One atom can register a transition; an ensemble spectrum supports abundance, excitation, velocity, broadening, and opacity inferences.

The periodic table supplies stable local identities and a hierarchy of retained nuclear and chemical history. It does not assign different fundamental gravity to different elements. Missing wavelengths and unresolved blends are Transmission limits; recorded unknown lines are Familiarity limits.

## 10 What a Picture of Jupiter Can and Cannot Say

A photograph supplies reflected or emitted light from weighted atmospheric depths. Spectroscopy identifies atoms and molecules. Thermal maps constrain heat flow. Microwave channels reach different depths. Gravity harmonics constrain mass distribution. Magnetic fields constrain conducting regions. In-situ samples provide local particle and isotope records.

No one channel contains the whole planet. Independent channels make different structures familiar while preserving honest degeneracies. The same logic applies to stars, galaxies, and black holes: a horizon-scale ring is reconstructed emission from plasma, not a photograph of an interior.

## 11 Reinterpreting a Working Simulation

For an accepted physical simulation $S(t + \mathrm{d}t) = \Phi_{\mathrm{d}t}[S(t)]$, UM acts downstream:

$$L_{\text{UM}}(t) = I_{\text{UM}}[\,S(t);\ \ell,\ \text{channels}\,].$$

The physical evolution, initial conditions, gravity, hydrodynamics, chemistry, and calibrated parameters remain unchanged. Every result is labelled as record **R**, UM interpretation **U**, or unresolved information **E**.

The supplied CAMB/Planck history produced: EXECUTED

| Established transition | Numerical location | UM interpretation |
|---|---|---|
| radiation–matter equality | $z = 3468.33$ | the two background records have equal weight |
| recombination | $z \approx 1089$ | the long-range CMB channel opens |
| accelerated expansion begins | $z = 0.6407$ | expansion overtakes matter deceleration |
| matter–$\Lambda$ equality | $z = 0.3022$ | the two density records are equal |

The radiation–matter crossing is resolution-stable to a fractional error of 0.0007076. Generic equality and mixing baselines find the same crossing. That validates the interpreter's calculation, but prevents the known transition from being claimed as unique UM evidence.

## 12 THE MECHANISM SECTORS AT A GLANCE

Under anchors A1–A4 and the calibration, the formal chain becomes a finite proposed mechanism. The routed cosmological readout, quantum registration, action lattice, compact carrier, and metallic branches are downstream tests of one invariant contraction $r$ — and their statuses differ.

| SECTOR | CONTENT | STATUS |
|---|---|---|
| Cosmological routing | unique routing matrix; $\Omega_b = \frac{r^2}{2}$, $\Omega_{\text{DM}} = 4r^2(1-r)$, $\Omega_{\text{DE}} = 1 - \frac{9}{2}r^2 + 4r^3$ under one common calibration | CONDITIONAL |
| Quantum registration | golden routing is a Born-compatible POVM with Perron frequencies $(1/\varphi, 1/\varphi^2)$; no automatic deviation | PROVED |
| Action lattice | coherence cell of volume $r^3$; one calibration $S_* r^3 = \hbar$ | CONDITIONAL |
| $E_8$ carrier and vacuum | rank-eight closed spin carrier; first shell 240; maximal-defect expectation $r^{240}$ read as $\rho_\Lambda / M_*^4$ | CONDITIONAL |
| Silver galactic branch | metallic spectrum $\mu_n^2 = n\mu_n + 1$ is exact; two-return galactic realization and rotation observable not yet derived | OPEN |
| Black-hole closure tree | age saturation, parent–child closure functor | PROPOSED |
| Hubble braid | $\text{Tr}(B_3) = 3r$ is a proved trace identity; the observation map $\Pi_H$ must be derived or measured | OPEN |

Cross-sector rigidity is the decisive unification criterion: on the golden branch, registration, baryons, dark matter, the action unit, and the vacuum hierarchy are one number, bound by $r$-free identities such as $h = 8b$, $q^2 = 8b^3$, $\lambda = (2b)^{120}$, and $(8b - d)^2 = 128\,b^3$. A framework that fails one identity and repairs it with a sector-specific parameter is, by construction, a different and larger theory.

## 13 THE 2026 EMPIRICAL PROGRAMME: WHAT WAS ACTUALLY TESTED

## 13.1 IllustrisTNG literature phase

EXECUTED

Because catalogue access was pending, a compute-light comparison was completed against nine primary TNG papers; twenty-six evidence statements were classified against the UM observational language: 14 strong retrospective mappings, 8 partial or compatible, 4 neutral, 3 recorded TNG-versus-observation tensions, and 0 direct UM contradictions.

> TNG LITERATURE VERDICT
>
> **Language survives; superiority untested.** This phase shows retrospective compatibility only. It does not establish that UM predicts the simulation or outperforms generic scientific language.

## 13.2 IllustrisTNG catalogue test

DEFERRED

A complete frozen execution package exists: reduced group and subhalo catalogues, hashed adapter and data, leakage prevention, spatially held-out partitions, and predeclared transfer runs (fit TNG100-1 at $z = 0$; apply unchanged to TNG100-2, TNG300-1, and an earlier epoch). The earlier adapter formula was not found in the available sources, so the adapter contract was deliberately left empty rather than guessed. The corrected adapter maps each galaxy independently, passes unit-rescaling tests to $4.44 \times 10^{-16}$ maximum error with zero catalogue-order error, and no population fraction is fixed in advance. The current TNG demonstration is an implementation test, not simulation evidence.

## 13.3 SPARC real-galaxy pilot

NULL RESULT

Twelve real SPARC galaxies (270 resolved radial measurements) were tested with leave-one-galaxy-out evaluation. Compiler shares: $a_L$ from the gas contribution, $a_M = 1 - a_L$ from the stellar contribution; UM coordinates $W_L = a_L^2$, $W_B = 2a_L a_M$, $W_M = a_M^2$.

| Model | Galaxy-balanced RMSE (dex) | Point RMSE (dex) | Velocity RMSE (km/s) |
|---|---|---|---|
| RAR-fixed | **0.2404** | **0.2582** | **17.15** |
| Generic linear inputs | 0.3154 | 0.3001 | 27.34 |
| UM quadratic coordinates | 0.3273 | 0.3133 | 30.60 |
| Generic quadratic baseline | 0.3273 | 0.3133 | 30.60 |
| Baryons only | 0.5275 | 0.5559 | 43.33 |

*Velocity column is galaxy-balanced RMSE. The fixed radial-acceleration relation was the best predictor; the UM and generic quadratic maps agreed to a maximum difference of* $1.421 \times 10^{-14}$ *dex.*

The agreement is algebraic, not accidental: $\{1, W_B, W_M\}$ and $\{1, a_L, a_L^2\}$ span the same quadratic feature space. The pilot therefore tested a translator/compiler, not a UM-derived galactic force law; the conditional silver-ratio branch was not tested, because its radial operator, baryonic coupling, and circular-velocity observable have not yet been derived.

> SPARC VERDICT
>
> **Real-data test completed. Compiler executed. No distinctive advantage found. The galactic mechanism remains open.** The result improves over baryons alone but does not beat the established RAR baseline, and is algebraically equivalent to a generic quadratic feature map in this implementation.

What this phase proved about roadblocks: the programme is not blocked by the impossibility of storing the whole physical world. It can perform serious tests on reduced, physically meaningful records. The actual bottleneck is always specific — an unavailable catalogue, a missing frozen adapter, or an observable that has not yet been derived — and must never be inflated into a claim that the framework is finished or untestable.

## 14 How the Core Can Succeed or Fail

The ensemble compiler must pass:

1. **resolution stability** — major labels and relations persist across justified coarse-graining scales;
2. **transfer** — unchanged rules work across snapshots, volumes, instruments, or systems;
3. **withheld prediction** — retained coordinates predict fields not used to construct them;
4. **shuffle null** — performance vanishes when record associations are permuted;
5. **generic baseline** — performance exceeds simpler density, threshold, mixing, clustering, or dimensional-reduction descriptions;
6. **honest uncertainty propagation** — noise, low $N_{\text{eff}}$, covariance, and missing channels widen the result;
7. **no dynamical inflation** — descriptive success is not reported as discovery of a new force.

If it does not beat generic baselines, UM can remain a coherent descriptive language without a distinctive empirical advantage — and that result must be published as such, exactly as the SPARC pilot was.

## 15 Optional New Physics

HYPOTHESIS H

The core framework needs no new field. A separate optional branch considers an even, density-gated conformal scalar with a symmetron-type potential. Its UM-specific restriction is the dilute visible/retained response ratio

$$q = \frac{\beta_C}{\beta_U} = \frac{r^2}{u} \approx 0.1381966011, \qquad \text{UU} : \text{UC} : \text{CC} \;=\; 1 : 0.1381966 : 0.0190983$$

in the unscreened quasistatic limit. The activation density, range, coupling amplitude, abundance, initial conditions, and vacuum offset are not derived by UM. After the density threshold is crossed, the field need not instantly occupy its dilute equilibrium value; activation by the present day depends on its range, cosmic history, initial fluctuations, and possible domain walls. Those are testable constraints, not details to assume away.

The branch must first survive stability, equivalence-principle, laboratory, Solar-System, nucleosynthesis, CMB, expansion, growth, and nonlinear-screening tests. Parameters are then fitted on declared training data and frozen before withheld lensing, velocity, gas, and environmental predictions. If the fixed ratio fails while a free-ratio model succeeds, the UM branch fails. That failure does not invalidate the record-to-field framework.

## 16 CLAIMS THIS MONOGRAPH DOES NOT MAKE

UM does not claim that:

- generic self-description uniquely forces the golden ratio;
- structural channel weights are automatically cosmic densities;
- three channel types prove three spatial dimensions;
- age alone creates geometry or a scalar field;
- powers of $r$ automatically predict cosmological or particle observables;
- periodic-table position creates element-specific gravity;
- $E_8$, 240 vacuum events, black-hole entropy, or a silver galaxy follow without additional bridge postulates;
- an image reveals a hidden interior;
- a retrospective compatibility result is a successful prediction;
- numerical proximity is empirical confirmation.

## 17 FINAL STATUS

UM is **complete** as a typed reconstruction framework, a language of scale-dependent identity and Boundary, a separation of absent from unfamiliar information, a four-clock age system, a periodic and cross-scale observational ledger, a record-to-field ensemble rule, an executable downstream interpreter of an accepted cosmological history, and a finite proposed mechanism bounded by four anchors and one calibration.

UM is **not yet** a validated replacement for established physics, a uniquely superior ensemble language on independent structure data, a theory of dark-matter abundance, a solution to vacuum energy, or an empirically confirmed golden coupling theory. The TNG literature phase shows language survival only; the SPARC pilot is an honest baseline-equivalence null for a translator; the galactic law and the observation map $\Pi_H$ remain open derivations.

The remaining work is external science: obtain catalogue access and the exact frozen adapter, run the structure test with transfer and withheld prediction against generic baselines, derive the missing observables, and separately determine whether the optional fixed scalar-response ratio survives existing constraints and independent data.

## Reference Anchors

1. NIST, *Periodic Table of the Elements*.
2. IUPAC, *Periodic Table of Elements*.
3. Manohar, *Effective Field Theories*.
4. Lin and Lu, *Mori–Zwanzig model reduction*.
5. Planck Collaboration, *Planck 2018 cosmological parameters*.
6. Lewis, Challinor and Lasenby, *CAMB lineage*.
7. Nelson et al., *IllustrisTNG public data release*.
8. Lelli, McGaugh and Schombert, *SPARC: mass models for 175 disk galaxies*.
9. Hinterbichler and Khoury, *Symmetron fields*.
10. Touboul et al., *MICROSCOPE final result*.
11. Llinares and Pogosian, *Symmetron domain walls*.
12. Event Horizon Telescope Collaboration, *M87* shadow and mass*.

End Note

The central idea is simple: the mathematics does not need to be a new substance. It can be the exact language of what many physical records collectively describe, at the scale where that description becomes stable and useful — and it becomes physics only where a mechanism, an observable, a record, and a failure condition are named.

One Invariant · Many Sectors · One Consistency Test

$\varphi$